

Arizona State University
School of Electrical, Computer and Energy Engineering

EEE 465/591: Photovoltaics Energy Conversion
Practice Problem Set #7

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1. Battery Capacity Calculation

Calculate the required battery capacity (in Ah) for a 12V system to power a 60W refrigerator for 14 hours at night, considering Lead-Acid batteries can only be discharged to 50% capacity (50% DoD).

Energy Consumed at Night:

$$E_{\text{night}} = 60 \text{ W} \times 14 \text{ h} = 840 \text{ Wh}$$

Total Battery Capacity Needed (accounting for 50% DoD):

$$E_{\text{total}} = 840 \text{ Wh} / 0.50 = 1680 \text{ Wh}$$

Ampere-hours at 12V:

$$\text{Ah} = 1680 \text{ Wh} / 12 \text{ V} = 140 \text{ Ah}$$

Answer: **140 Ah**

2. Required Panel Power Generation

Determine the power needed from solar panels to both run the refrigerator and recharge the battery during 10 hours of sunlight.

Energy Required to Recharge (assuming 80% charging efficiency):

$$E_{\text{recharge}} = 840 \text{ Wh} / 0.80 = 1050 \text{ Wh}$$

Recharge Power (over 10 sunlight hours):

$$P_{\text{recharge}} = 1050 \text{ Wh} / 10 \text{ h} = 105 \text{ W}$$

Total Panel Power (Load + Recharge):

$$P_{\text{total}} = 60 \text{ W (Load)} + 105 \text{ W (Recharge)} = 165 \text{ W}$$

Answer: **165 W**

3. Bonus: Feasibility Check

Determine if a standard c-Si panel (96 cells, $V_{\text{mp}} = 54.2\text{V}$, $I_{\text{mp}} = 5.45\text{A}$) can run the refrigerator and charge the battery, assuming 10 hours at maximum power point and 95% converter efficiency.

Given Panel Specifications:

- $V_{mp} = 54.2 \text{ V}$
- $I_{mp} = 5.45 \text{ A}$
- $V_{oc} = 63.3 \text{ V}$
- $I_{sc} = 5.83 \text{ A}$
- Converter efficiency = 95%

Panel Output Power (P_{mp}):

$$P_{mp} = 54.2 \text{ V} \times 5.45 \text{ A} = 295.39 \text{ W}$$

Useful Power (after converter loss):

$$P_{\text{useful}} = 295.39 \text{ W} \times 0.95 = 280.62 \text{ W}$$

Conclusion:

Since $P_{\text{useful}} = 280.62 \text{ W} > P_{\text{required}} = 165 \text{ W}$, the panel provides sufficient power.

Answer: Yes, the panel can support the system

Summary:

Parameter	Value
Panel Maximum Power	295.39 W
Useful Power (95% efficiency)	280.62 W
Required Power	165 W
Excess Power Available	115.62 W